Title (125 characters max)

- **A (104):** Closing 130 photogrammetric meshes of brain dissections: watertight is easy, trustworthy volumes are not
- **B (108):** Photogrammetry of brain dissections: automatic mesh closing works, but acquisition limits volume measurement

Impact Statement (40 words max)

Photogrammetric meshes of brain dissections can be closed automatically, but their scanned edge moves by millimetres between epochs, changing volumes as much as a dissection step: acquisition must be controlled before these meshes serve as quantitative references.

Synopsis (100 words max)

Motivation: Photogrammetric meshes of brain dissections could serve as quantitative references for diffusion MRI, but they are open surfaces with scanning defects.

Goal(s): Close them into watertight volumes and measure how far these volumes can be trusted.

Approach: We repaired and closed all 130 meshes of the public BraDiPho archive, compared closing methods and checked that volume decreases along each dissection.

Results: Every mesh became watertight, and closing methods changed volumes by 0.45% on average. Yet volume increased in 15 of 104 dissection steps, by up to 15%, mostly because the scanned edge moved 2 to 5 mm outward.

Introduction

Photogrammetry records a brain dissection as a series of textured 3D meshes, one per step ("epoch")[1,2]. The BraDiPho archive[1] shares such meshes to compare dissected pathways with diffusion MRI tractography, and surface scans of dissected brains already support quantitative measures[3]. Measuring the volume removed at each step requires watertight surfaces. However, each mesh is open on the face that was not photographed and contains scanning defects (inverted faces, folds, floating debris), and a watertight mesh is not necessarily a correct one (Figure 1).

We closed the whole archive and asked: (1) can the meshes be closed automatically; (2) how much does the volume depend on how they are closed; and (3) do the volumes behave as a dissection should? We report what works, what does not, and what we wish we had known at the start.

Methods

Data. We used all 130 epochs of the 12 BraDiPho specimens[1]. Undocumented "b" variants were excluded from comparisons between epochs, leaving 104 consecutive steps.

Repair and closing (Python, VTK; Figure 2). (1) Duplicated faces, small flaps and small fragments are removed, non-manifold edges are split, and faces are oriented outward. (2) Each opening, bounded by a loop, is closed by a cap. A plane is fitted to the loop; if the loop projects onto it without crossing itself (after straightening small folds of up to 20 vertices), a planar cap is triangulated[4] on the outer side of the opening. Otherwise, a curved cap is built by harmonic interpolation. (3) Remaining defects and debris are removed, and each part is oriented outward. (4) Volume is computed only for watertight meshes.

Evaluation. Baselines: a simple fan closing of the same loops, two variants of fold straightening (A: always planar; B: planar only if the cap adds at most 5% of volume) and MeshFix[5], an independent

repair and hole-filling method. We measured (a) watertightness; (b) the closing uncertainty, i.e. the volume difference between methods; (c) visual defects in every epoch; and (d) plausibility: dissection only removes tissue, so volume should decrease. Each increase was compared with 7 normal steps for scale (similarity ICP), displacement of the main cap and volume added by the caps.

Results

What works. All 130 epochs became watertight (Figure 2). Straightening small folds reduced the epochs needing a curved cap from 38 to 12. Volumes differed from the fan closing by 0.45% on average (maximum 3.4%) and from MeshFix by [TODO: mean, max over 130 epochs; 0.73% and 2.2% on the first 21] (Figure 3c). After corrections, the epochs with visual defects dropped from 29 to 12.

What works poorly. Most of these 12 epochs (7 from one specimen) have truly folded openings, where the curved cap folds onto itself (Figure 4a); others have small holes just under the planar cap (Figure 4b). On four deeply hollowed openings, variants A and B differed by 5.7 to 11% in volume: there, the volume depends on an arbitrary choice.

What does not work. The median volume removed per step was 8.7% (range 0.5–37.5%). Yet volume *increased* in 15 of 104 steps, by up to 15.3%, and 21 changes were smaller than the closing uncertainty (Figure 3a–b). Scale differences between epochs ($\leq$0.8%) did not explain the increases. In 9 of the 15, including the three largest, the main cap lay 2.2 to 5.1 mm further out, adding 19 to 57 cm^3, whereas it moved by -2.2 to $+1.9$ mm in normal steps (Figure 5). Two others followed a poorly closed epoch, and four were within the closing uncertainty. [TODO: Check that MeshFix shows the same 15 increases.]

Discussion

A watertight mesh is not necessarily correct: a three-vertex fold can warp a planar cap, and inverted debris silently subtracts volume; only visual inspection revealed them. The closing uncertainty (about 3%, up to 11%) cannot explain increases of 13 to 15%: the scanned edge itself moves, so any closing method inherits the shift. Each millimetre changes the volume by 2 to 4%, comparable to a typical dissection step. The limiting factor is the acquisition, not the closing. We recommend photographing both faces at every epoch, fixing the specimen on a rigid support with markers and a scale reference, reprocessing the photographs with identical settings, and reporting the closing uncertainty with every volume. Without reference volumes, real tissue removal on the unphotographed face cannot yet be separated from settling or movement of the specimen.

Figure captions

Preview figure

Two options (one is submitted):

- **a)** "Closed, but not trustworthy": left, a closed epoch with its caps in red (*130/130 watertight*); right, two consecutive epochs overlaid (*+15% volume while tissue was removed*).
- **b)** One closed hemisphere, caps in red, with three short lines: *closing: ±0.5%*, *scanned edge: ±2 mm = ±5–8%*, *fix the acquisition first.*

References (Vancouver, in order of first citation)

1. Vavassori L, Rheault F, Nocerino E, et al. Brain dissection photogrammetry: a tool for studying human white matter connections integrating ex-vivo and in-vivo multimodal datasets. Nat Commun.

figures/fig1_concept.png (missing)

Figure 1: *

Figure 1 (`fig1_concept.png`, 416 characters). Why a watertight volume can still be wrong. Cross-section of a dissected hemisphere (grey), open on the unphotographed face. (a) The cap (red) follows the edge of the scanned surface. (b) If this edge lies 1 mm further out at the next epoch, the volume grows by 2–4%, as much as a typical dissection step. (c) A three-vertex fold of the boundary warps a planar cap. (d) Inverted debris silently subtracts its volume.

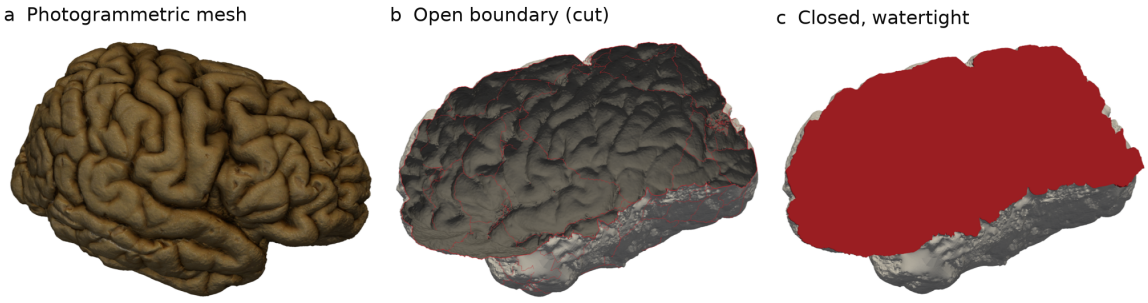


Figure 2: *

Figure 2 (`fig1_pipeline.png`, 209 characters). Pipeline on a successful epoch (spc-01, epoch 1). (a) Textured photogrammetric mesh. (b) Open boundary, seen from the unphotographed face (inside of the shell in dark grey). (c) Watertight result, caps in red.

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2. De Benedictis A, Nocerino E, Menna F, et al. Photogrammetry of the human brain: a novel method for three-dimensional quantitative exploration of the structural connectivity in neurosurgery and neurosciences. World Neurosurg. 2018;115:e279–e291. doi:10.1016/j.wneu.2018.04.036

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4. Shewchuk JR. Triangle: engineering a 2D quality mesh generator and Delaunay triangulator. In: Lin MC, Manocha D, editors. Applied Computational Geometry. Lect Notes Comput Sci. 1996;1148:203–222. doi:10.1007/BFb0014497

5. Attene M. A lightweight approach to repairing digitized polygon meshes. Vis Comput. 2010;26:1393–1406. doi:10.1007/s00371-010-0416-3

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Institutional logo

[TODO: Université de Sherbrooke logo]

Keywords (5)

Ex-Vivo Applications; Validation; Precision & Accuracy; Analysis/Processing; Brain.

Authors and categories (ECHO form)

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- **Abstract type:** Standard.
- **Primary category:** Analysis Methods – Data Processing.
- **Secondary category:** Diffusion – Simulation/Validation.

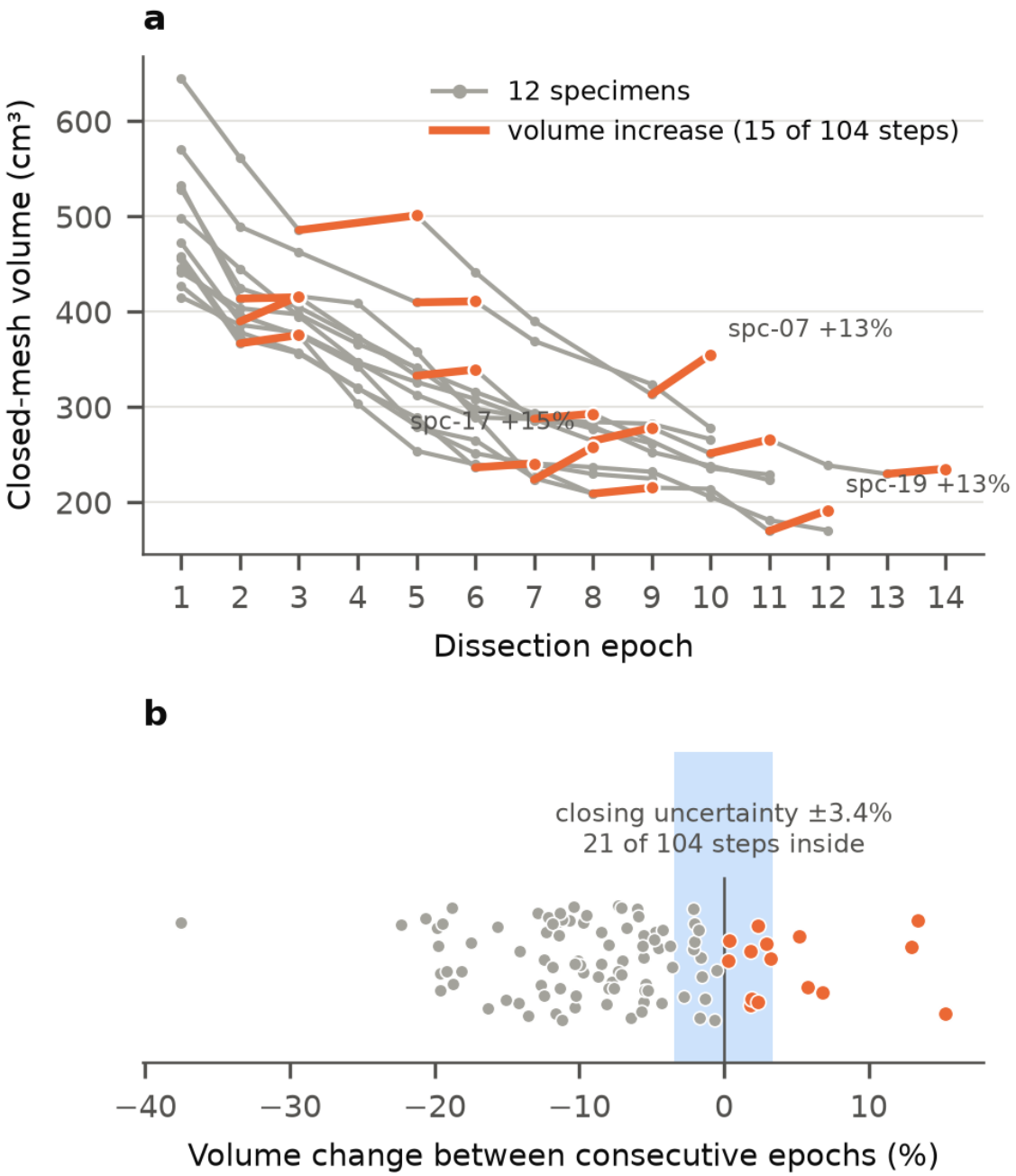


Figure 3: *

Figure 3 (`fig2_volume_vs_epoch.png`, 370 characters). Volumes of the 12 specimens. (a) Closed volume versus dissection epoch; volume should only decrease, orange steps increase (15 of 104, up to 15%). (b) Volume change of each step; blue band: closing uncertainty (±3.4%), which contains 21 steps. (c) Closing uncertainty of each epoch: volume difference between our closing and a fan closing, variants A and B, and MeshFix. [TODO: add panel (c)]

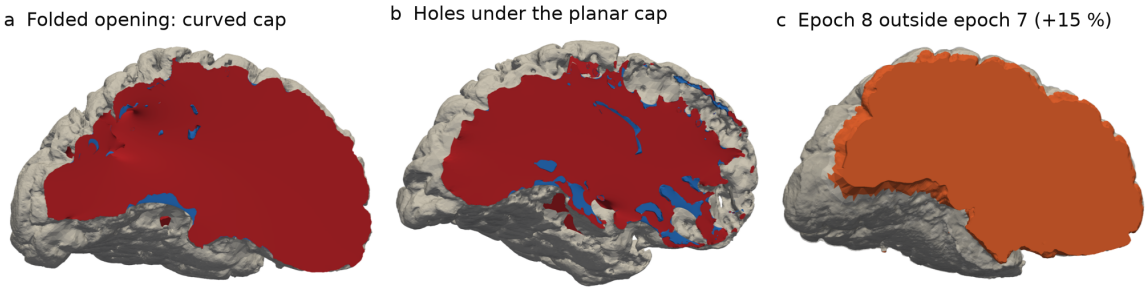


Figure 4: *

Figure 4 (`fig3_failures.png`, 183 characters). What works poorly (caps in red, back-facing surfaces in blue). (a) Folded opening (spc-07, epoch 2): the curved cap folds. (b) Small holes just under the planar cap (spc-15, epoch 8). [TODO: move panel (c) to Figure 5]

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Figure 5: *

Figure 5 (`fig5_scanned_edge.png`, 333 characters). Where the volume increases come from. (a) Epoch 8 (orange) lying outside epoch 7 (grey) of spc-17: the scanned edge moved 4.1 mm outward and the volume rose by 15%. (b) Outward displacement of the main cap versus volume change, for the 15 increases (orange) and 7 normal steps (grey). Dashed lines: 2 and 4% of volume per millimetre.